

Matthieu Blanke

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RESEARCH POSITIONS

NYU Courant Institute of Mathematical Sciences **New York, USA**
Postdoctoral Associate **2025**
Deep learning for climate, advised by Sara Shamekh and Pierre Gentine.

EDUCATION

Inria Paris and DI ENS **Paris, France**
Ph.D. **2021-2024**
Deep learning for physical systems, advised by Marc Lelarge.

École Normal Supérieure Paris-Saclay **Paris, France**
Master of Science *Mathématiques, Vision, Apprentissage (MVA)* **2020-2021**
Machine learning, optimization and statistics. Also passed exams of statistical physics courses at ENS Paris.

École polytechnique **Paris, France**
Engineer's degree *Cycle ingénieur polytechnicien* **2017 - 2021**
Applied mathematics, computer science, theoretical physics

WORK EXPERIENCE

Inria Paris **Paris, France**
Research internship **April - September 2021**
Deep implicit layers with applications to physical systems. Advised by Marc Lelarge.

Econophysix **Paris, France**
Research internship **June - September 2020**
Particle-based stochastic modeling of the latent order book. Advised by Michael Benzaquen and Jean-Philippe Bouchaud.

Saildrone **Alameda, CA, USA**
Platform team intern **June - August 2019**
Processing of oceanographic data collected by drones, fault detection.

PUBLICATIONS

An updated list is available on [my Google Scholar page](#).

- Matthieu Blanke, Yongquan Qu, Sara Shamekh, Pierre Gentine. Strictly Constrained Generative Modeling via Split Augmented Langevin Sampling. Submitted. 2025
- Matthieu Blanke, Ronan Fablet, Marc Lelarge. Neural Incremental Data Assimilation. Accepted at the AI for Science workshop, at ICML 2024.
- Matthieu Blanke, Marc Lelarge. Interpretable Meta-Learning of Physical Systems. The Twelfth International Conference on Learning Representations, 2024 ([ICLR 2024](#)). Also presented at the Synergy of Scientific and Machine Learning Modeling workshop, at ICML 2023 (SynS & ML 2023).

- Matthieu Blanke, Marc Lelarge. FLEX: an Adaptive Exploration Algorithm for Nonlinear Systems. The Fortieth International Conference on Machine Learning, 2023 ([ICML 2023](#)). Also presented at the Machine Learning and the Physical Sciences workshop, at NeurIPS 2022.
- Matthieu Blanke, Marc Lelarge. Online greedy identification of linear dynamical systems. 61st Conference on Decision and Control, 2022 ([CDC 2022](#)). Also presented at the Adaptive Experimental Design and Active Learning in the Real World workshop, at ICML 2022.
- Matthieu Blanke, Jose Moran, Pierre-Philippe Crépin, Jean-Philippe Bouchaud, Michael Benzaquen. Market impact in a multiple metaorder landscape. Under submission.

PROJECTS AND SOFTWARE

The projects are available on [my GitHub page](#).

Physics exercise book	Ongoing
Open source exercise book for undergraduate students preparing for the competitive exams for the top French engineering schools.	
Neural Incremental Data Assimilation	2024
Deep learning model for the data assimilation inverse problem, built with JAX.	
CAMEL	2023
Interpretable meta-learning of physical systems	
FLEX	2022
Adaptive exploration of physical systems.	
Deep Latent Variable Models	2021
Maximum likelihood sampling and missing data imputation based on deep learning. Supervised by S. Allassonnière.	
Market impact simulator	2020
Python module for noisy market impact experiments.	
Image segmentation: Random Walker and SegNet	2019
Theoretical study and Python implementation of the Random Walker algorithm. Performance benchmark versus the deep learning architecture SegNet. Supervised by S. Allassonnière.	
Automatic sport scene modeling	2020
A C++ library that automatically detects the players' positions on a sport video.	
Machine learning for power consumption forecast	2019
Clustering, pattern detection and statistics on weather data from Météo France to predict energy consumption. Implemented the algorithms from scratch in C++.	
IP-over-Discord	2018
A C program and a Node.JS Discord Bot for IP-over-Discord network tunneling.	

TALKS

06/2025	M²LiNES annual meeting, invited talk	New York
Physically Constrained Generative Modeling.		
02/2025	Columbia LEAP center, invited talk	New York
Interpretable Meta-Learning of Physical Systems.		
09/2024	Huawei Lavender Summit 2024, invited talk	Bordeaux

Interpretable Meta-Learning of Physical Systems.

03/2024 **Mines Paris Geosciences Department, invited talk** Paris
Interpretable Meta-Learning of Physical Systems.

07/2023 **MLIA seminar, invited talk** Paris
Interpretable Meta-Learning of Physical Systems.

12/2023 **ICML in Paris, contributed talk** Paris
FLEX: an Adaptive Exploration Algorithm for Nonlinear Systems.

04/2023 **Oral presentation for Safran Research** Paris
Exploration of physical systems.

12/2022 **ML4Physical Sciences workshop, poster presentation** Paris
Online exploration of nonlinear physical systems.

10/2022 **Inria PhD Seminar, oral presentation** Paris
Exploration of physical systems.

06/2022 **GdR IASIS, poster presentation** Paris
Online greedy identification of physical systems. "Apprentissage et modélisation physique" workshop.

09/2021 **CIRM Workshop, poster presentation** Marseille
Deep learning isochronism. "On Future Synergies for Stochastic and Learning Algorithms" workshop.

TEACHING

Mines PSL Paris, France
Teaching assistant Fall 2023
Probability theory. Differential equations.

Université Paris Cité Paris, France
Teaching assistant Spring 2023
Numerical physics.

Université Paris 1 Panthéon-Sorbonne Paris, France
Teaching assistant Fall 2022
Statistics.

Université Paris Cité Paris, France
Teaching assistant Spring 2022
Numerical physics.

Université Paris 1 Panthéon-Sorbonne Paris, France
Teaching assistant Fall 2021
Statistics.

École polytechnique, X-Talents Paris, France
Oral Examiner 2018
Weekly mathematics and physics oral tests preparing undergraduate students for the competitive entrance examinations of French top engineering schools.

SERVICE

Organizer of Inria Argo team's seminar 2022-2024

Reviewing NeurIPS 2025, ICLR 2025, IEEE Control Systems Letters 2024, IEEE CDC 2024, ICML Workshop Synergy of Scientific and Machine Learning Modeling 2023, NeurIPS workshop on Machine Learning and the Physical Sciences 2022

SCHOLARSHIPS

2025 - 2027 Google.org ["AI for societal impact" scholarship](#)

2021 École polytechnique's AMX PhD Scholarship

SKILLS

Computer languages Python, Julia, C, C++, Node.js, $\text{\LaTeX}$

Python frameworks PyTorch, JAX

LANGUAGES

Frenc Native

Italian Native

English Fluent

German Advanced

Spanish Conversational